The transhipment problem

Standard form

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Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Standard form

The transhipment problem is a linear optimization problem.
The simplex algorithm that we have seen requires the problem to be written in standard form.
This is what we are doing in this video.

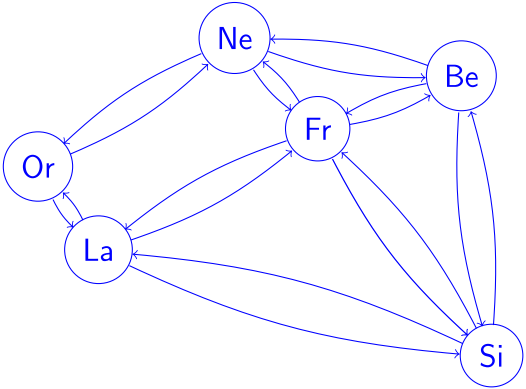
Linear optimization problem

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

$$\sum_{j|(i,j)\in\mathcal{A}} x_{ij} - \sum_{k|(k,i)\in\mathcal{A}} x_{ki} = s_i, \, \forall i \in \mathcal{N}.$$

$$\ell_{ij} \leq x_{ij} \leq u_{ij}, \, \forall (i,j) \in \mathcal{A}.$$



Set the lower bounds to zero

$$x'_{ij} = x_{ij} - \ell_{ij} \; x_{ij} = x'_{ij} + \ell_{ij}$$

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x'_{ij} + \sum_{(i,j) \in \mathcal{A}} c_{ij} \ell_{ij} = \min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x'_{ij}$$

subject to

$$\sum_{j|(i,j) \in \mathcal{A}} x'_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x'_{ki} = s_i + \sum_{k|(k,i) \in \mathcal{A}} \ell_{ki} - \sum_{j|(i,j) \in \mathcal{A}} \ell_{ij} = s'_i, \; \forall i \in \mathcal{C},$$

$$0 \leq x'_{ij} \leq u_{ij} - \ell_{ij} = u'_{ij}, \; \forall (i,j) \in \mathcal{A}.$$

Slack variables

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x'_{ij}$$

subject to

$$\sum_{j | (i,j) \in \mathcal{A}} x'_{ij} - \sum_{k | (k,i) \in \mathcal{A}} x'_{ki} = s'_i \; \forall i \in \mathcal{C},$$

$$0 \leq x'_{ij} \leq u'_{ij}, \forall (i,j) \in \mathcal{A}.$$

Slack variables

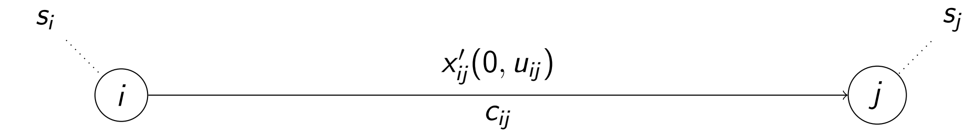
$$0 \le x'_{ij} \le u_{ij}, \ \forall (i,j) \in \mathcal{A}.$$

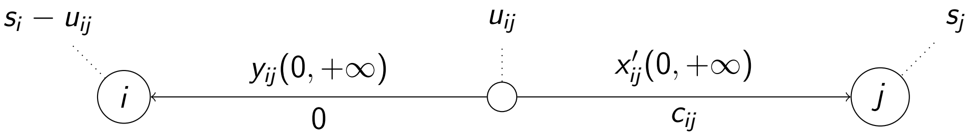
Slack variables: y_{ij}

$$x'_{ij} + y_{ij} = u_{ij},$$
$$x'_{ij} \ge 0, \ y_{ij} \ge 0.$$

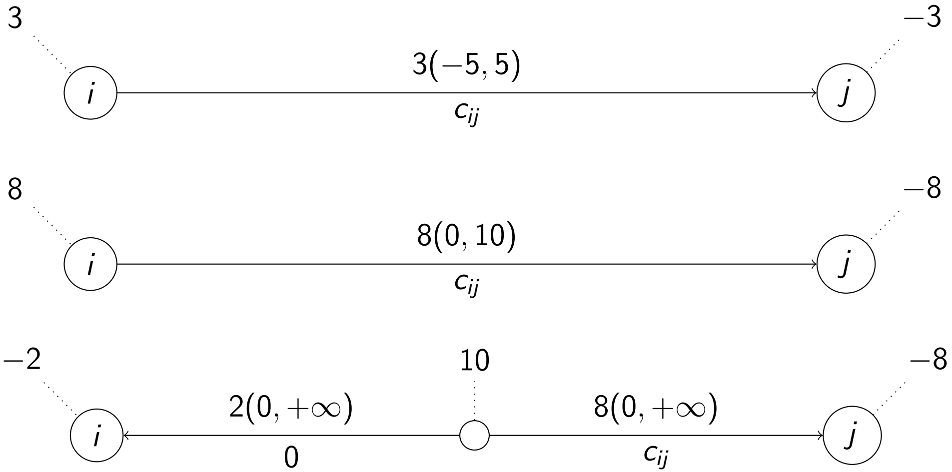
Interpretation

$$x'_{ij} + y_{ij} = u_{ij}$$





Illustration



Problem in standard form: modified network

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

$$\sum_{j | (i,j) \in \mathcal{A}} x_{ij} - \sum_{k | (k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N},$$

$$x_{ij} \geq 0, \forall (i,j) \in \mathcal{A}.$$

$$\min_{x \in \mathbb{R}^n} c^T x$$

subject to

$$Ax = s,$$

$$x \geq 0.$$

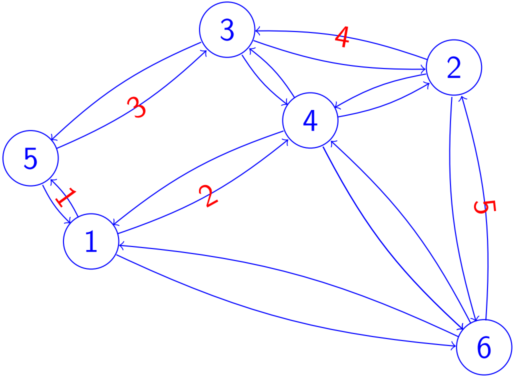
Incidence matrix

$$\sum_{j|(i,j)\in\mathcal{A}} x_{ij} - \sum_{k|(k,i)\in\mathcal{A}} x_{ki} = s_i, \ \forall i \in \mathcal{N}.$$

For each arc (i, j) of index k,

$$a_{ik} = 1 \text{ and } a_{jk} = -1.$$

Incidence matrix



	1	2	3	4	5	…
1						
2						
3						
4						
5						
6						

Summary

The standard form can be interpreted as a transhipment problem on a modified network

The matrix defining the flow conservation constraints is called the incidence matrix

It has a specific structure: the column for arc (i, j) contains exactly two non zero element: a 1 at row i and a -1 at row j.